

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTMC503 - THEORY OF MACHINES- II

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Due to Slip of the Belt, the efficiency of the belt drive
a) Decreases b) Increases c) Does not change d) none of the Options
Belt drives are used for
a) Change of Speed b) change of Torque c) Change of Speed & Torque both d) None of the Options
- 2 Module of the Gear is the ratio of
a) Number of Teeth/ Pitch Circle Diameter b) Pitch Circle Diameter / Number of Teeth c) Pitch Circle Radius / Number of Teeth d) None of the Options
- 3 In case of Cross Belt Drives, the pulley rotates in
a) Same direction b) Opposite direction c) Same or Opposite direction d) None of the Options
- 4 Following tooth profiles obeys the Law of Gearing
a) Involute b) Cycloidal c) Both Involute & Cycloidal d) None of the Options
- 5 In case of damped vibrations, the amplitude of vibration..... w.r.t time.
a) Increases b) Decreases c) Remains Constant d) None of the Options
- 6 In forced vibrations the frequency of the vibration is frequency of excitation force.
a) More than b) Less than c) Equal to d) None of the options
- 7 The flywheel is used to
a) Store the energy b) Reduce fluctuation in speed c) Both options a & b d) None of the options
- 8 The rotor of a ship rotates in clockwise direction when viewed from the stern and the ship takes a left turn. The effect of the gyroscopic couple acting on it will be
a) to raise the bow and stern b) to lower the bow and stern c) to raise the bow and lower the stern d) to lower the bow and raise the stern
- 9 The train value of a gear train is
a) equal to velocity ratio of a gear train b) reciprocal of velocity ratio of a gear train c) always greater than unity d) always less than unity

- 11 In a simple gear train, if the number of idle gears is odd, then the motion of driven gear will
 a) be same as that of driving gear b) be opposite as that of driving gear c) depend upon the number of teeth on the driving gear d) none of the options
- 12 The maximum fluctuation of energy is the
 a) sum of maximum and minimum energies b) difference between the maximum and minimum energies c) ratio of the maximum energy and minimum energy d) ratio of the mean resisting torque to the work done per cycle
- Q2. Solve the following.
- A) Explain the phenomena of 'slip' and 'creep' in a belt drive ? 6
- B) Find the power transmitted by a flat belt running over a pulley of 600 mm diameter at 200 r.p.m. The coefficient of friction between the belt and the pulley is 0.25, angle of lap 160° and maximum tension in the belt is 2500 N.
- Q3. Solve the following.
- A) Power is transmitted using a V-belt drive. The included angle of V-groove is 36° . The belt is 25 mm deep and mean width is 20 mm. If the mass of the belt is 0.3 kg per meter length and maximum allowable stress is 1.5 MPa, determine the maximum power transmitted when the angle of lap is 120° . $\mu = 0.17$.
- B) State & explain the advantages & disadvantages of Helical Gears over Spur Gears.
- Q4. Solve Any Two of the following.
- A) Explain in brief various types of gear trains with neat sketches. 6
- B) A pair of parallel helical gears consists of a 20 teeth pinion meshing with a 60 teeth gear. The helix angle is 15° and the normal pressure angle is 20° . The normal module is 4 mm.
 Determine 1) Transverse module 2) Pitch circle diameters of the pinion and the gear 3) Center distance.
- C) In an epicyclic gear train, an arm carries two gears A and B having 36 and 45 teeth respectively. If the arm rotates at 150 r.p.m. in the anticlockwise direction about the center of the gear A which is fixed, determine the speed of gear B.
- Q5. Solve Any Two of the following.
- A) The mass of flywheel of an engine is 6.5 tonnes and the radius of gyration is 1.8 metres. It is found from the turning moment diagram that the fluctuation of energy is 56 kN-m. If the mean speed of the engine is 120 r.p.m., find the maximum and minimum speeds.
- B) Explain the following terms related to flywheel 6
 a) Fluctuation of Speed b) Coefficient of Fluctuation of Speed c) Coefficient of Fluctuation of Energy
- C) An aeroplane makes a complete half circle of 50 metres radius, towards left, when flying at 200 km per hr. The rotary engine and the propeller of the plane has a mass of 400 kg and a radius of gyration of 0.3 m. The engine rotates at 2400 r.p.m. clockwise when viewed from the rear. Find the gyroscopic couple on the aircraft and state its effect on it.
- Q6. Solve Any Two of the following.

- A) Explain the following terms related to vibrations 6
a) Forced Vibrations b) Damped Vibrations c) Critical or Whirling Speed of Shaft
- B) Explain the following types of free vibrations. 6
a) Longitudinal Vibrations b) Transverse Vibrations c) Torsional Vibrations
- C) A shaft 50 mm diameter and 3 metres long is simply supported at the ends and carries three loads of 1000 N, 1500 N and 750 N at 1 m, 2 m and 2.5 m from the left support. The Young's modulus for shaft material is 200 GN/m^2 . Find the frequency of transverse vibration.

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